

# Literature Status: Energy-Concentrated Wave-Packet Frames on Higher-Dimensional Euclidean Balls

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**Status: literature already answered.** The higher-dimensional problem recorded after Theorem 1.2 of arXiv:2601.21224 is answered by the same authors in the separate later paper arXiv:2607.23953. Its Theorem 1.2 constructs the requested frame for every  $d \geq 2$  and obtains the desired  $R^{d-1}$  bound up to a polylogarithmic factor.

## 1 Original question

K. Hughes, A. Israel, and A. Mayeli, *Wave Packets and Eigenvalue Estimates for Limiting Operators on the Disk*, arXiv:2601.21224 [1], proves the relevant energy-concentration theorem for  $d = 2$ . Immediately after Theorem 1.2, on arXiv PDF page 3, the authors identify the remaining case as follows.

**Source open problem.** For  $d \geq 3$ , construct a wave-packet frame on the  $d$ -dimensional Euclidean ball whose index set decomposes as  $\mathcal{I} = \mathcal{I}_1 \cup \mathcal{I}_2 \cup \mathcal{I}_3$ , with

$$\sum_{\nu \in \mathcal{I}_1} \|\hat{\psi}_\nu\|_{L^2(S)}^2 + \sum_{\nu \in \mathcal{I}_2} \|\hat{\psi}_\nu\|_{L^2(\mathbb{R}^d \setminus S)}^2 \leq \varepsilon^2,$$

and with residual cardinality

$$\#\mathcal{I}_3 \leq CR^{d-1} \log(R/\varepsilon)^J$$

for some  $J > 0$  and  $C > 0$ .

The source states explicitly that its theorem answers this problem for  $d = 2$  and that dimensions  $d \geq 3$  remain open.

## 2 Separate later answer

The same authors subsequently wrote *Localized frames on Euclidean balls*, arXiv:2607.23953 [2]. Its introduction explicitly describes the paper as extending the disk framework to Euclidean balls in arbitrary dimensions. Theorem 1.2, on arXiv PDF pages 2–3, supplies the exact requested construction.

For the codimension-one case  $\eta = 1$ , specialize the theorem’s geometric error term to obtain the following.

**Later theorem (specialized to codimension one).** Fix  $d \geq 2$  and  $s > 1$ . For every dyadic  $R \geq 2$  there is a unit-norm frame  $\{\psi_\nu\}_{\nu \in \mathcal{I}}$  for  $L^2(B_d(R))$ , with frame bounds depending only

on  $d$ . For every eligible bounded frequency set  $S$  and  $\varepsilon \in (0, 1/2]$ , its index set has the energy decomposition in the source question and

$$\#\mathcal{I}_3 \lesssim_{d,s} \mathcal{M}^{d-1}(\partial S) R^{d-1} \log \left( \frac{R \operatorname{diam}(S)}{\varepsilon} \right)^{sd+1}.$$

After the standard normalization of  $\operatorname{diam}(S)$ , this is the required estimate with  $J = sd + 1$ . The dyadic restriction is removed by the rescaling noted directly after the theorem. Thus the later paper answers all dimensions  $d \geq 3$  from the source problem. It is stronger in another direction as well: its full statement treats boundaries of fractional codimension  $\eta \in (0, 1]$ .

### 3 Status and bounded search

A bounded search through 9 August 2026 checked the run registry and cheap local indexes for arXiv:2601.21224, followed by arXiv searches using the exact title, authors, and the phrases “wave packet frame”, “Euclidean ball”, and “higher dimensions”. The decisive match was arXiv:2607.23953. Its theorem and its explicit relation to the disk paper were checked in both the arXiv source and PDF. This packet records an already-known later answer and claims no new mathematical result.

### References

- [1] K. Hughes, A. Israel, and A. Mayeli, *Wave Packets and Eigenvalue Estimates for Limiting Operators on the Disk*, arXiv:2601.21224 (2026).
- [2] K. Hughes, A. Israel, and A. Mayeli, *Localized frames on Euclidean balls*, arXiv:2607.23953 (2026).